Lesson2 Reflection of Light

- Law of reflection
- Periscope
- Regular and diffuse reflection
- Practical work: Reflection by a plane mirror

If we know how light behaves when it is reflected, we can use a mirror to change the direction in which the light is travelling. This happens when a mirror is placed at the entrance of a concealed drive to give warning of approaching traffic.

An ordinary mirror is made by depositing a thin layer of silver on one side of a piece of glass and protecting it with paint. The silver – at the *back* of the glass – acts as the reflecting surface.

1.Law of reflection

Terms used in connection with refl ection are shown in Figure 27.1. The perpendicular to the mirror at the point where the incident ray strikes it is called the **normal**. Note that the **angle of incidence** *i* is the angle between the incident ray and the normal; similarly the **angle of reflection** *r* is the angle between the refl ected ray and the normal.

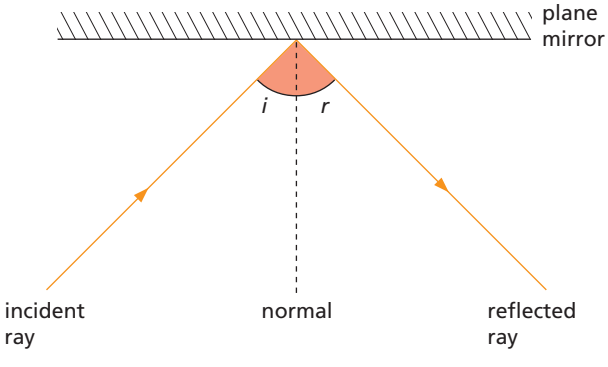


Figure 27.1 Reflection of light by a plane mirror

The law of reflection states:

> The angle of incidence equals the angle of reflection.

The incident ray, the reflected ray and the normal all lie in the same plane. (This means that they could all be drawn on a flat sheet of paper.)

Practical work

Reflection by a plane mirror

Draw a line AOB on a sheet of paper and using a protractor mark angles on it. Measure them from the perpendicular ON, which is at right angles to AOB. Set up a plane (flat) mirror with its reflecting surface on AOB.

Shine a narrow ray of light along, say, the 30° line onto the mirror (Figure 27.2).

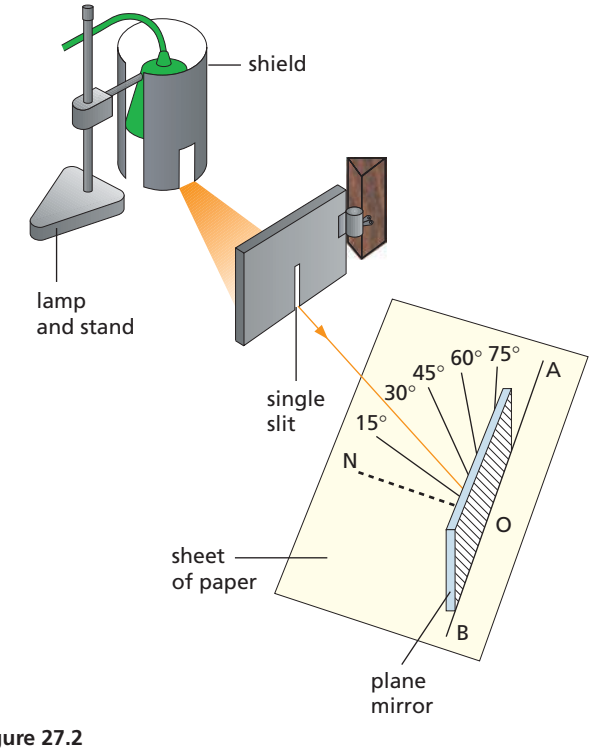


Figure 27.2

Mark the position of the reflected ray, remove the mirror and measure the angle between the reflected ray and ON. Repeat for rays at other angles. What can you conclude?

2. Periscope

A simple **periscope** consists of a tube containing two plane mirrors, fixed parallel to and facing each other. Each makes an angle of 45° with the line joining them (Figure 27.3). Light from the object is turned through 90° at each reflection and an observer is able to see over a crowd, for example (Figure 27.4), or over the top of an obstacle.

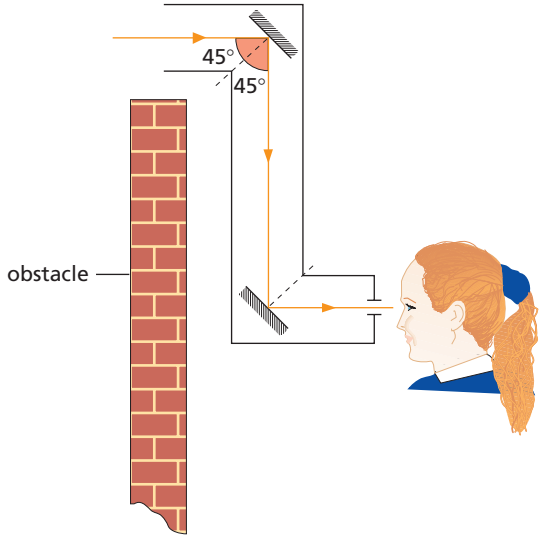


Figure 27.3 Action of a simple periscope



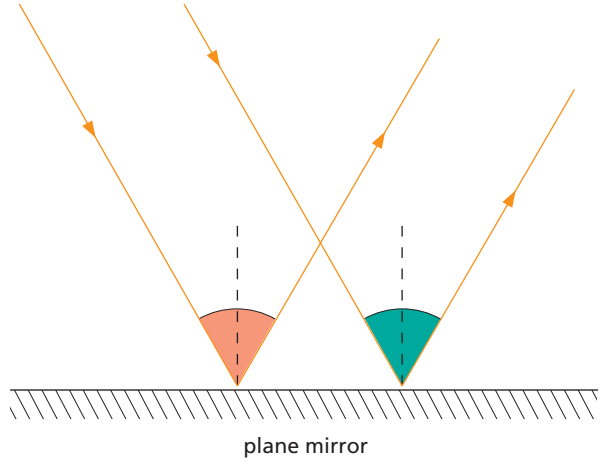
Figure 27.4 Periscopes being used by people in a crowd.

In more elaborate periscopes like those used in submarines, prisms replace mirrors (see Chapter 30).

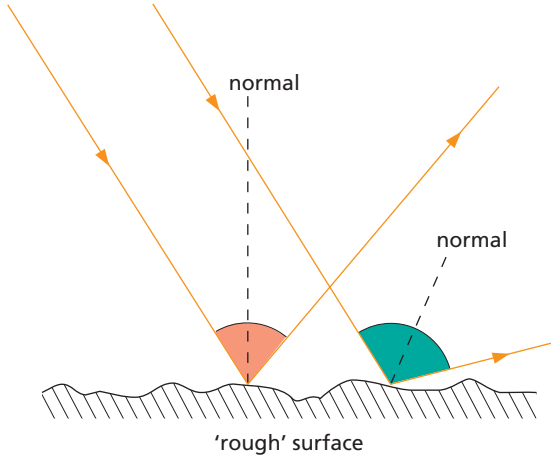
Make your own periscope from a long, narrow cardboard box measuring about 40 cm × 5 cm × 5 cm (Such as one in which aluminium cooking foil or clingfilm is sold), two plane mirrors (7.5 cm × 5 cm) and sticky tape. When you have got it to work, make modifications that turn it into a 'see-back-o-scope', which lets you see what is behind you.

3. Regular and diffuse Reflection

If a parallel beam of light falls on a plane mirror it is reflected as a parallel beam (Figure 27.5a) and **regular reflection** occurs. Most surfaces, however, reflect light irregularly and the rays in an incident parallel beam are reflected in many directions (Figure 27.5b).



a Regular reflection



b Diffuse reflection Figure 27.5

Irregular or **diffuse reflection** happens because, unlike a mirror, the surface of an object is not perfectly smooth. At each point on the surface the laws of reflection are obeyed but the angle of incidence, and so the angle of reflection, varies from point to point. The reflected rays are scattered haphazardly. Most objects, being rough, are seen by diffuse reflection.

Checklist

After studying this chapter you should be able to

- state the law of reflection and use it to solve problems,
- describe an experiment to show that the angle of incidence equals the angle of reflection,
- draw a ray diagram to show how a periscope works.

Questions

1 Figure 27.6 shows a ray of light PQ striking a mirror AB. The mirror AB and the mirror CD are at right angles to each other. QN is a normal to the mirror AB.

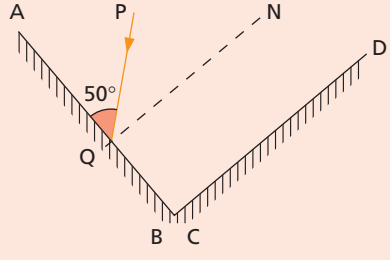


Figure 27.6

a What is the value of the angle of incidence of the ray PQ on the mirror AB?
b Copy the diagram, and continue the ray PQ to show the path it takes after reflection at both mirrors.
c What are the values of the angle of reflection at AB, the angle of incidence at CD and the angle of reflection at CD?
d What do you notice about the path of the ray PQ and the final reflected ray?

2 A ray of light strikes a plane mirror at an angle of incidence of 60°, is reflected from the mirror and then strikes a second plane mirror placed so that the angle between the mirrors is 45°. The angle of reflection at the second mirror, in degrees, is
A 15 **B** 25 **C** 45 **D** 65 **E** 75

3 A person stands in front of a mirror (Figure 27.7). How much of the mirror is used to see from eye to toes?



Figure 27.7